GROUP 5 · IIT MADRAS DS AND AI LAB · 2026 T3 · MILESTONE 2

SerpentGuard AI — Dataset, SAM Masks & EDA

Offline Indian Snake Segmentation, Classification & Safety Decision Support (**86 / 86 Automated Checks Passed**)

PROCESSED DATASET	TAXONOMY & TIERS	WHO MEDICAL PRIORITY	SAM MASKS & INTEGRITY
181,231	**208 Classes**	**17 WHO Classes**	**129,241 Masks**
VERIFIED SNAKE PHOTOS	**68 TIER-1 · 140 TIER-2**	**6 CAT 1 · 11 CAT 2 CLASSES**	**100% TRAIN SAM COVERAGE**
95,787 sightings by 25,041 photographers	+ 81 Unknown Pool species (656 photos)	22,969 photos (12.7% of dataset)	0 shared observers/hashes across splits

Three Attached Kaggle Datasets & Presentation Scope

I. DATASET ARCHITECTURE & LABELS

- `serpentguard-indian-snakes-processed` (181,231)
- `serpentguard-m2-splits` & Cleaning Funnel
- Long-Tail Imbalance (Gini 0.76) & 14 Families
- WHO SEARO 2016 Category 1 & 2 Coverage

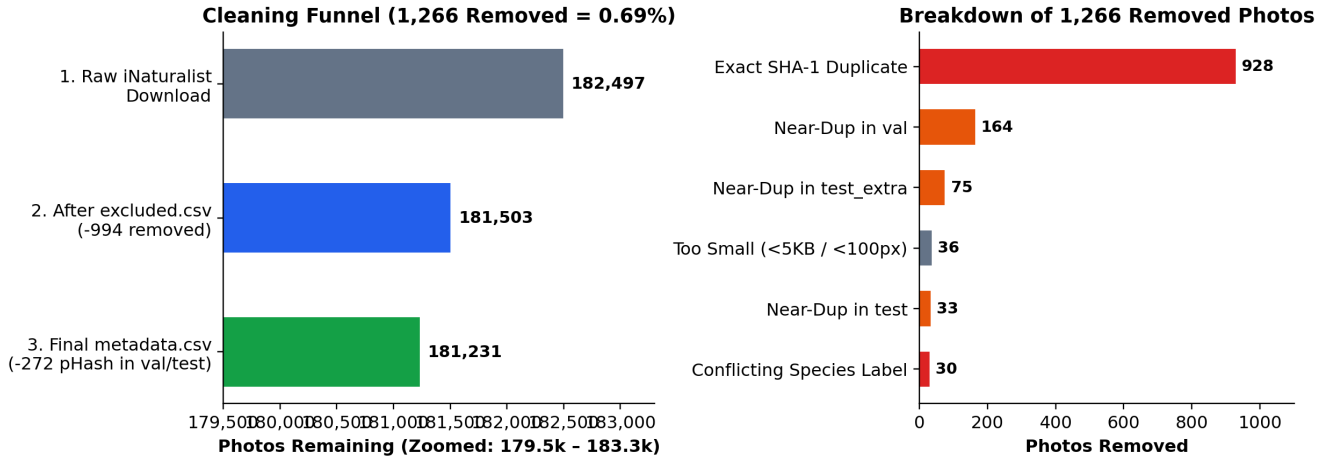
II. SPLITS, COVARIATE SHIFTS & PIXEL AUDIT

- Observer-Grouped Splits & 2022+ India Test Rule
- Hidden Shifts: Portrait Flip & Country Shortcut
- Snake Box Size at 224/384 px & Non-JPG Outliers
- 6,784-Photo Pixel Quality & Custom RGB Norms

III. EMBEDDINGS, LOOKALIKES & SAM MASKS

- DINOv2 Embeddings (Domain AUC ~0.50 & Cleanlab)
- 9 M1 Lookalike Pairs + Unlisted Venomous Relatives
- `serpentguard-sam-masks` (129,241 Masks Audit)
- Geography, Licences & Milestone 3 Action Plan

From 182,497 Raw Downloads to 181,231 Verified Photos Across 3 Kaggle Datasets

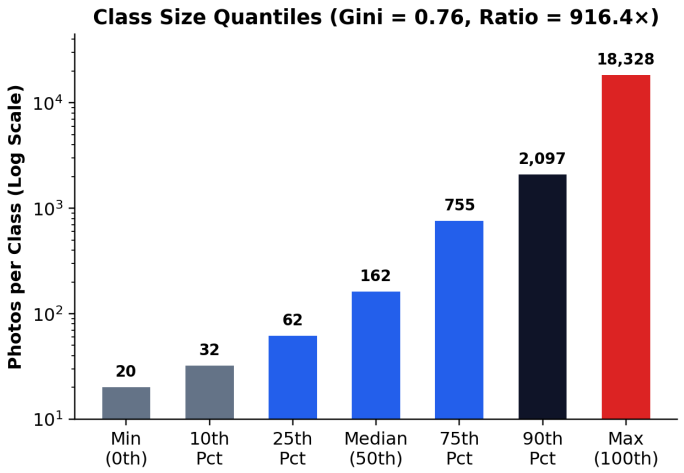


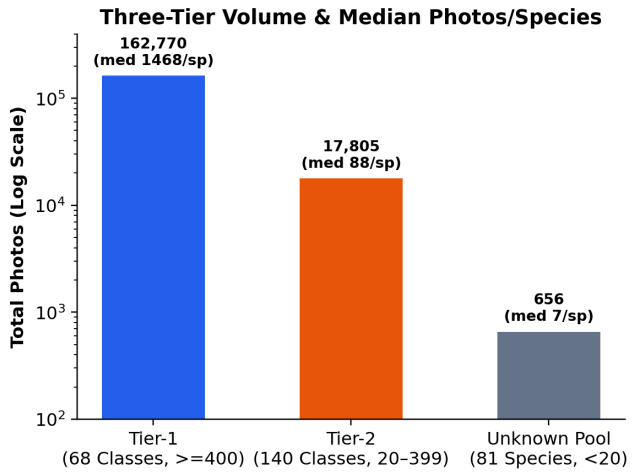
Two-Stage Cleaning & Disk Verification

- **Stage 1 Hygiene (`excluded.csv` = 994 rows)**: Removed **928** exact SHA-1 duplicates, **30** conflicting-species uploads, and **36** broken/tiny files (<5 KB: 19; <100 px: 17) → **181,503** rows in `splits.csv`.
- **Stage 2 Split Deduplication (-272 rows)**: Dropped cross-split pHash duplicates from `val` (-164), `test_extra` (-75), and `test` (-33) without touching `train` → **181,231** final rows.
- **100% Disk & Schema Match**: Every row in `metadata.csv` (33 raw columns) matches an image on disk; only `latitude/longitude` have **1,488** obscured gaps (0.82%).

KAGGLE DATASET HANDLE	CORE FILES & DIMENSIONS	VERIFIED INTEGRITY CHECK
`serpentguard-indian-snakes-processed`	`metadata.csv` (181,231 × 33) · `classes.csv` (208 × 13) · `processing_report.json`	0 missing files on disk · 0 byte mismatches · 0 SHA-1 errors
`serpentguard-m2-splits` & `sam-masks`	`splits.csv` (181,503 × 31) · `excluded.csv` (994 × 25) · `mask_stats.csv` (129,241 × 17)	100% of `train` photos (129,241) linked to a precomputed SAM mask

Severe Long-Tail Skew (Gini = 0.76, Ratio = 916.4×): 13 Classes Hold 50% of Photos



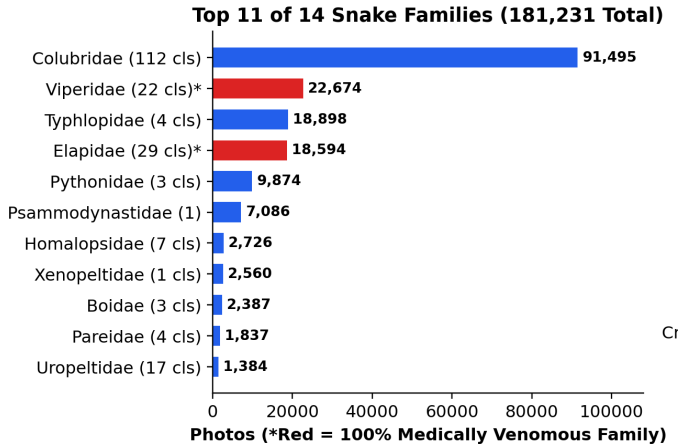


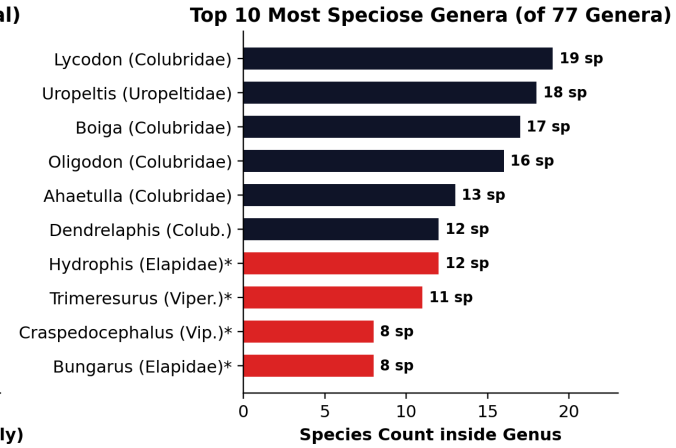
Quantifying the Imbalance

- **Effective Classes = 32.8 (vs. 208 Real)**: Top 10 classes hold **45.0%** of all class photos (180,575), while the 50 smallest classes hold just **1.0%** (80 classes have <100 photos).
- **Three-Tier Split**: **Tier-1** (68 classes, ≥400 photos) holds **162,770 photos** (median 1,468/sp); **Tier-2** (140 classes, 20–399) holds **17,805 photos** (median 88/sp).
- **Open-Set Holdout & M1 Core-27**: **81 rare species** (≤19 photos; 656 total) form `unknown_pool`; all **27 M1 Core species** sit in Tier-1 (**77,044 photos** = 42.7%).

IMBALANCE METRIC	ALL SPLITS (208 CLASSES)	TRAIN ONLY (129,241 PHOTOS)	MODELING IMPLICATION
Largest / Median / Smallest Class	18,328 / 161.5 / 20 photos	13,311 / 117.0 / 13 photos	Plain accuracy reflects only the top ~13 classes; must track macro-F1 & per-tier recall
Imbalance Ratio & Effective Classes	**916.4×** · Gini **0.76** · $N_{\text{eff}} = 32.76$	**1,023.9×** · Gini **0.76** · $N_{\text{eff}} = 32.48$	Requires tempered class-balanced sampling ($1/N_c^{0.5}$) and logit-adjusted focal loss

Colubridae Holds 50.5% of Photos; Viperidae & Elapidae Form 41,268 Venomous Photos



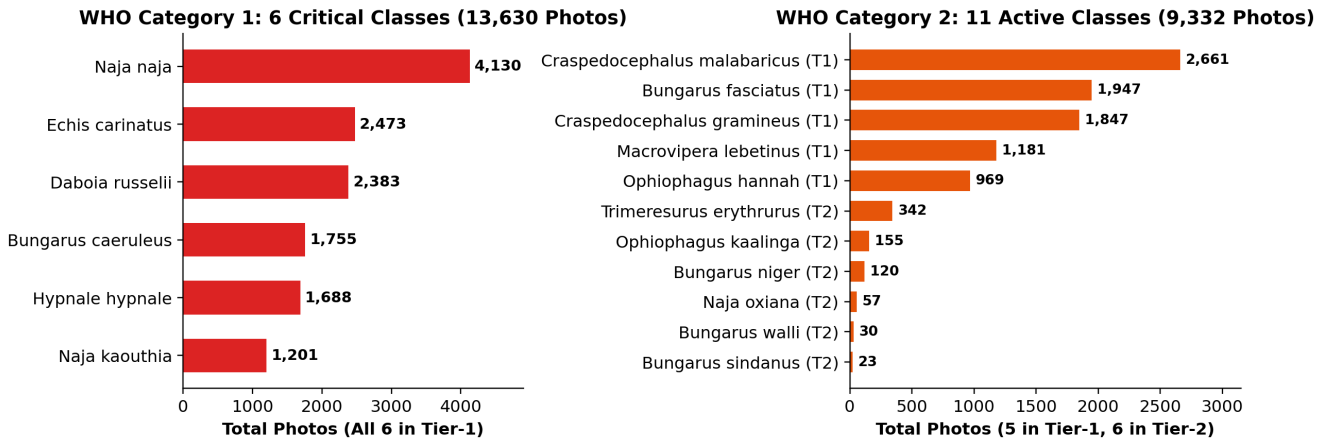


Clade Structure & Subspecies

- **Family Concentration**: **Colubridae** dominates with **91,495 photos** (148 species, 112 classes, 33 genera). At the tail, *Xenodermidae* (5 photos, 1 sp) sits entirely in `unknown_pool`.
- **100% Venomous Families (22.8%)**: **Viperidae** (22,674 photos, 22 classes) and **Elapidae** (18,594 photos, 29 classes) are medically venomous across every species.
- **Subspecies & Synonyms (4.7%)**: **8,578 photos** carry a subspecies/synonym in `record_taxon` (e.g. *C. helena helena* 959, *E. carinatus sochureki* 793) folded into the 208 species.

TAXONOMIC LEVEL	COVERAGE IN DATASET	WHY IT MATTERS FOR MULTI-TASK CLASSIFICATION
14 Families	Colubridae (91.5k), Viperidae (22.7k), Typhlopidae (18.9k), Elapidae (18.6k)	Family head separates 100% venomous Elapids/Vipers from harmless Typhlopids/Colubrids
77 Genera (29 Monotypic)	*Lycodon* (19 sp, 5,816), *Uropeltis* (18 sp, 825), *Boiga* (17 sp, 5,405), *Oligodon* (16 sp, 3,703)	Coarse genus prediction provides a reliable fallback for rare 20-photo Tier-2 species

17 Active WHO Classes (22,962 Photos): All 6 Category 1 Species Exceed 1,200 Photos

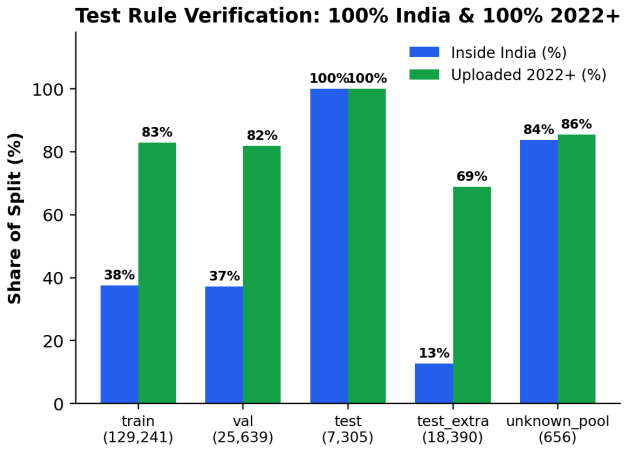


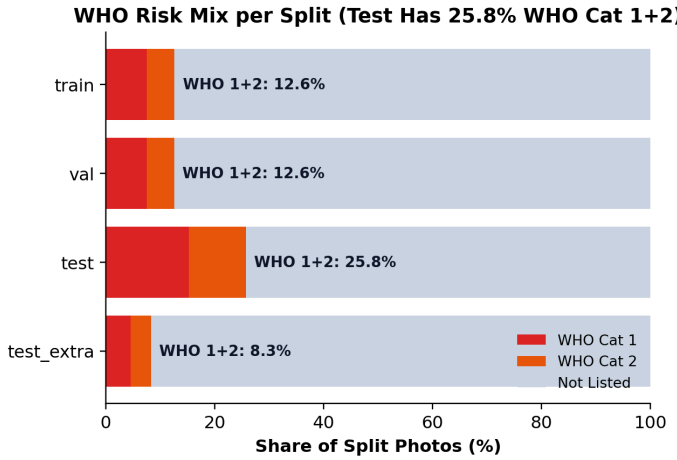
Medical Priority Breakdown

- **WHO Category 1 (6 Classes — 13,630 Photos, 7.5%)**: Highest mortality/morbidity. Median **2,069 photos/class**; all 6 sit in Tier-1 (*Naja naja* 4,130 down to *Naja kaouthia* 1,201).
- **WHO Category 2 (12 Species, 11 Classes — 9,339 Photos, 5.2%)**: Median **342 photos/class**; 5 in Tier-1, 6 in Tier-2, plus 1 ultra-rare Andaman species (7 photos) in `unknown_pool`.
- **Not Listed (271 Species, 191 Classes — 158,262 Photos, 87.3%)**: Median **151 photos/class**; prevents unnecessary panic and antivenom waste.

WHO CATEGORY	SPECIES / ACTIVE CLASSES	PHOTOS (SHARE)	MEDIAN PHOTOS / CLASS	TEST SPLIT COVERAGE WARNING
Category 1 (Critical)	6 species · 6 classes (all Tier-1)	13,630 (7.5%)	**2,069.0**	All 6 classes have 24–385 India 2022+ test photos
Category 2 (Important)	12 species · 11 classes (5 T1, 6 T2)	9,339 (5.2%)	**342.0**	**3 classes have 0 test photos** (*M. lebetinus*, *N. oxiana*, *B. walli*) → evaluate on `val/test_extra`

Zero Shared Observers/Hashes Across Splits & 100% India 2022+ Test Set



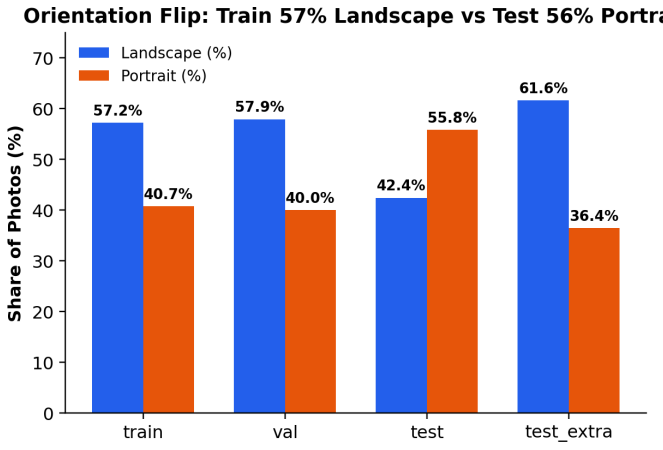


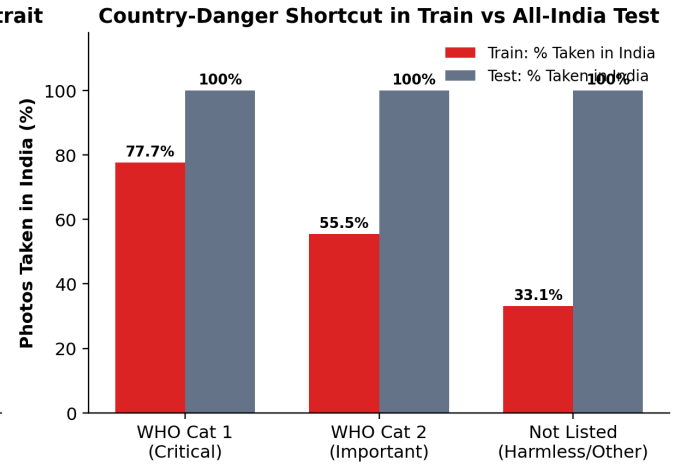
Strict Grouping & Test Rule

- **Zero Leakage Across Train, Val & Test**: Verified **0 shared** `observer_id`, `obs_id`, `sha1`, and exact `phash` across all 3 evaluation splits (67.9% of photos belong to multi-photo sightings).
- **100% India & 2022+ Test Rule (0 Violations)**: All **7,305 `test` photos** were taken in India and uploaded ≥2022-01-01 (vs. ~37.5% India in `train`/`val`).
- **28 Classes Have 0 Test Photos**: Due to the strict India 2022+ filter, 28 of 208 classes (including 3 WHO Cat 2 species) have `test = 0` and must be scored on `val` / `test_extra`.

SPLIT PARTITION	PHOTOS (SHARE)	IN INDIA %	UPLOADED 2022+ %	WHO CAT 1 + CAT 2 SHARE
`train` · `val`	129,241 (71.3%) · 25,639 (14.1%)	37.6% · 37.2%	82.9% · 81.8%	12.63% (7.54% + 5.09%) · 12.59% (7.57% + 5.02%)
`test` · `test_extra`	7,305 (4.0%) · 18,390 (10.1%)	**100.0%** · 12.7%	**100.0%** · 68.9%	**25.76%** (15.26% + 10.50%) · 8.32% (4.52% + 3.80%)

Four Hidden Shifts: Portrait Flip in Test, Country Shortcut & Thin Test Observers



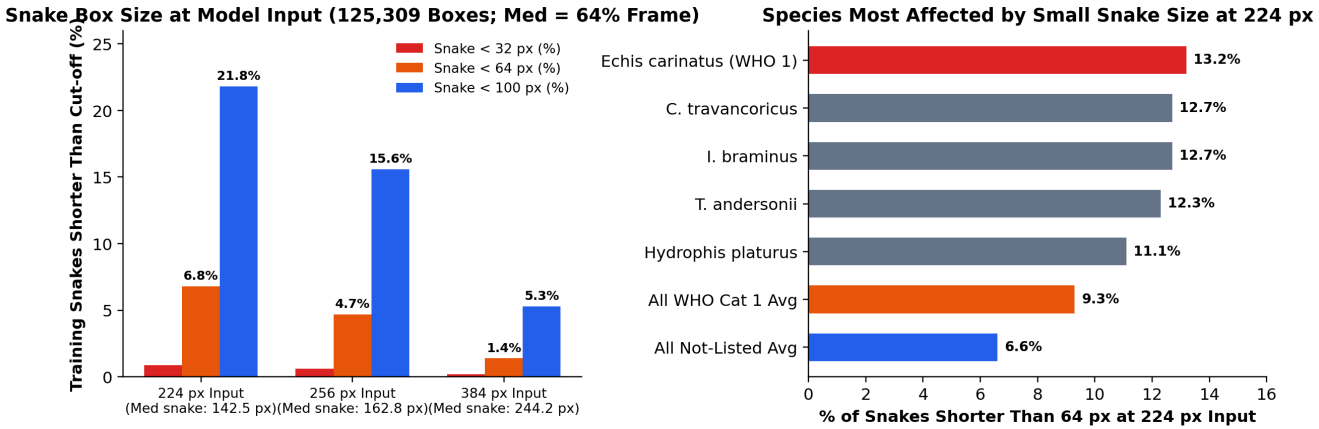


Why Deeper Checks Matter

- **Orientation Flip**: `train` is **57.2% landscape**, whereas `test` is **55.8% portrait** (recent Indian mobile uploads) → requires aspect-preserving padding and portrait validation tracking.
- **Country-Danger Shortcut in Train**: In `train`, **77.7% of WHO Cat 1** photos are from India vs. only **33.1% of Not Listed**. Background scenery is a false hint that fails on 100%-India `test`.
- **Near-Copies & Thin Test Observers**: **19** val/test photos sit within Hamming ≤2 bits of train (only **1** in `test`); **82 of 180** test classes have ≤3 test photographers (vs. only 3 of 208 in train ≤5).

HIDDEN AUDIT DIMENSION	MEASURED FINDING IN NOTEBOOK	MITIGATION FOR MILESTONE 3
pHash Near-Copies (1–2 bits)	0 identical (0-bit) across splits; 13 in `val`, 1 in `test`, 5 in `test_extra` within ≤2 bits	Purge the 19 ≤2-bit near-copies + 1 DINOv2 duplicate pair before freezing weights
Test Photographer Concentration	82 of 180 test classes have ≤3 photographers (10 classes have 1 photo from 1 person)	Compute test confidence intervals via photographer-grouped bootstrap, not per-photo

At 224 px Input, 6.8% of Snakes Are <64 px Long (13.2% for Echis carinatus)

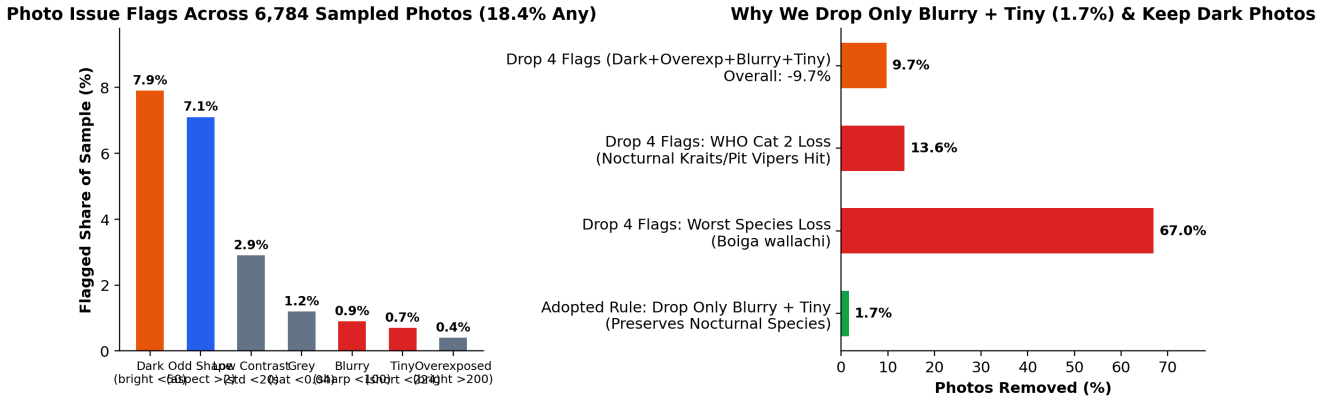


Resolution & Foreground Scale

- **500 px Medium Standard (24.4 GB Total)**: **99.66%** of photos have a long side of exactly 500 px (median 500×375 px, 125 KB). Only **0.67%** have short side <224 px.
- **Snake Box Length (125,309 Train Boxes)**: Median snake spans **64% of the photo's long side** (142.5 px at 224 input). However, **9.3% of WHO Cat 1 snakes** shrink below 64 px at 224 px (led by *Echis carinatus* at **13.2%**).
- **384 px or Box-Crop Fixes Small Snakes**: Raising input to 384 px cuts <64 px snakes from **6.8% → 1.4%**; cropping to the detector box eliminates the issue entirely.

FILE & HOST AUDIT	QUANTIFIED FINDING	DATALOADER ENGINEERING RULE
536 Non-JPG URLs & 26.2 MB Outliers	Top 100 bytes/pixel outliers are 62 PNGs & 35 GIFs saved with `.jpg` extension (max 26.2 MB)	Always call `.convert("RGB")` (fixes 18 PNGs/alpha & 1 greyscale found in 6,784 file audit; 0 EXIF rotations)
Two Hosts = Licence Split	S3 bucket (68.7% = 99.998% CC-licensed) vs. `static.inaturalist.org` (31.3% = ARR); Mann-Whitney $P = 0.488$	Zero quality or compression difference between CC and All-Rights-Reserved photos (124 vs 127 KB)

100% pHash Integrity, 0.1% Faces & Why We Keep Dark Nocturnal Snake Photos

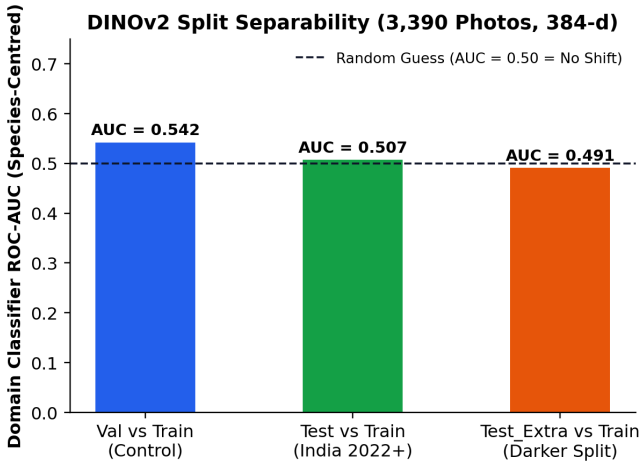


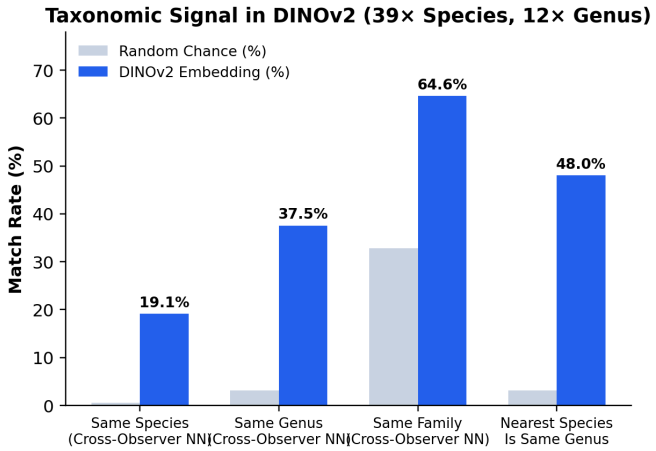
Pixel Quality & Cleaning Rule

- **6,784 Photos Measured (163 s, 0 Read Errors)**: **100.0%** match stored pHash (Hamming 0) and metadata dimensions; YuNet detects human faces in just **0.1%** of photos.
- **Why Not Drop Dark Photos (7.9%)?** Dropping dark + overexposed + blurry + tiny removes **9.7%** overall, hits **WHO Cat 2 hardest (13.6%)**, and wipes out **67%** of nocturnal species like *Boiga wallachi*.
- **Custom Train RGB Normalisation**: Our training set is darker than ImageNet defaults by `[-0.029, -0.017, -0.031]`, and `test_extra` is darker by up to `-0.231` SD in blue.

CHANNEL / QUALITY METRIC	TRAIN SAMPLE (N = 4,402)	IMAGENET DEFAULT	PREPROCESSING ACTION
RGB Mean & Std	Mean `[0.456, 0.439, 0.375]` · Std `[0.235, 0.224, 0.225]`	Mean `[0.485, 0.456, 0.406]` · Std `[0.229, 0.224, 0.225]`	Normalise with our own full-train RGB stats + keep brightness/contrast jitter for darker splits
Selective Cleaning Rule	Drop only **blurry** (<100 Laplacian) + **tiny** (<224 px) = **1.7% removed**	Preserves nocturnal kraits, wolf snakes & pit vipers	Pad odd aspect ratios (>2:1 = 7.1% overall, 10.9% in `test`) instead of cropping

No Hidden Domain Shift (Test AUC = 0.507) & Strong Clade Signal (12× Genus Match)



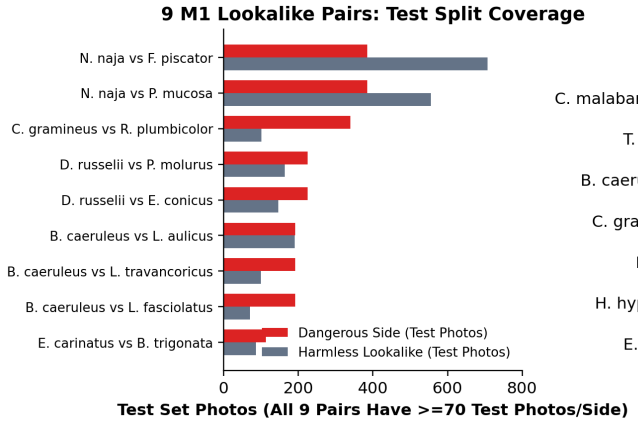


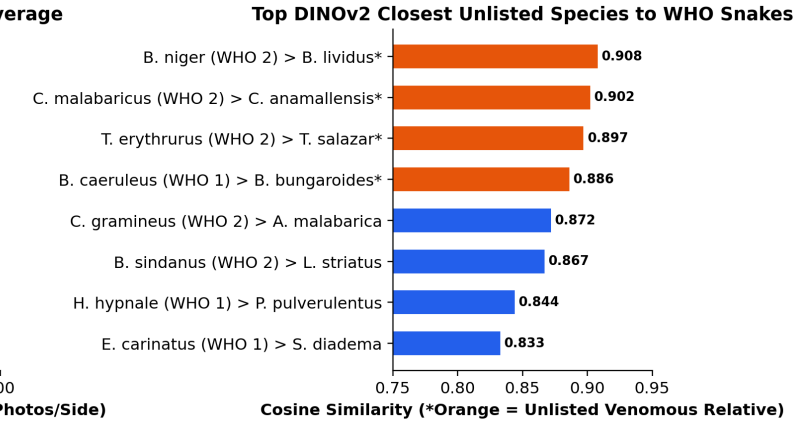
What DINOv2 Reveals

- **Splits Look Visually Indistinguishable**: A species-centred, observer-grouped classifier separating `train` from `val`, `test`, and `test_extra` gets **AUC = 0.542, 0.507, and 0.491** (≈0.50 random guess).
- **Test Coverage Matches Val**: Median cosine similarity to the closest same-species training photo is **0.569 in `test`** vs. **0.562 in `val`** and **0.574 in `test_extra`**.
- **Strong Hierarchical Signal**: Cross-photographer nearest neighbours match the **same species 19.1%** (39× chance), **same genus 37.5%** (12× chance), and **same family 64.6%**.

EMBEDDING AUDIT TASK	QUANTIFIED RESULT	ACTIONABLE FOLLOW-UP
Cross-Observer Near-Duplicates	14 photos at cosine ≥0.90; **1 pair at cosine 1.000 across `test` & `val`** (*Platyceps ladacensis*) missed by pHash	Inspect and remove the 14 high-similarity cross-observer pairs before locking final test evaluation
Cleanlab Confident Learning	Ranked `label_issue_candidates.csv`: 15.2% cross WHO categories; 12 match known lookalike pairs	Top flags are tiny/night/occluded snakes rather than wrong labels → hand-verify cross-WHO & lookalike flags only

All 9 M1 Lookalike Pairs Have ≥70 Test Photos; Embeddings Reveal Unlisted Relatives



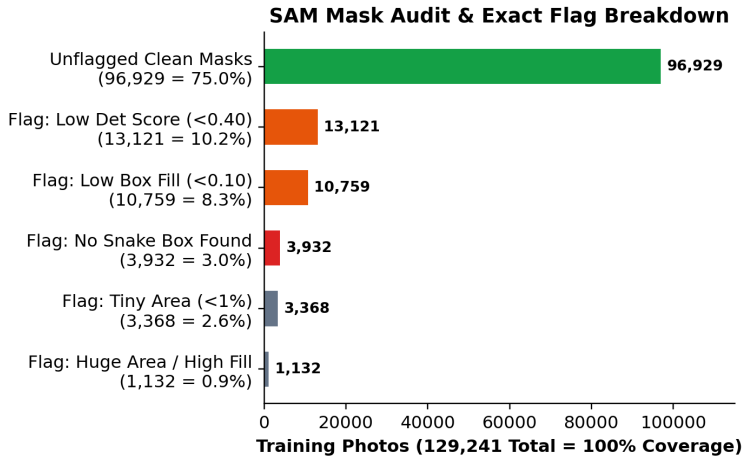


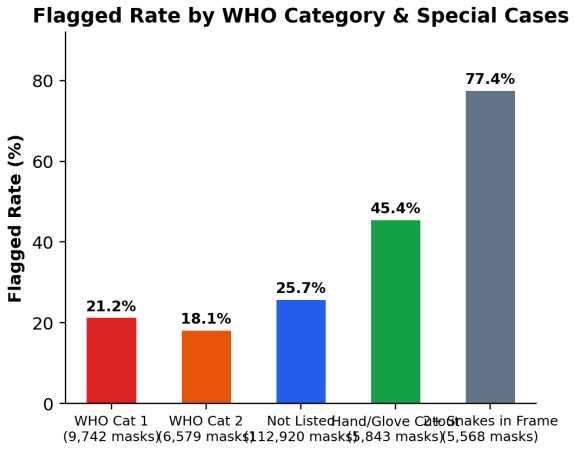
Beyond the 9 M1 Pairs

- **All 9 M1 Pairs Are Test-Ready**: Every pair has ≥70 test photos per side; in DINOv2 space, **6 of 9** dangerous snakes rank their named lookalike within the **top 5 closest species of 207**.
- **Sharpness Bias Warning**: *Bungarus caeruleus* photos are significantly sharper than *Lycodon aulicus* (median 2,496 vs 928, $p=0.006$) → blur augmentation is needed so models don't cheat via sharpness.
- **Critical Discovery ("Not Listed" ≠ Harmless)**: For 16 of 17 WHO species, the closest "not_listed" species in DINOv2 is a **venomous congener** (e.g. *Bungarus lividus* 0.908, *Craspedocephalus anamallensis* 0.902).

LOOKALIKE CATEGORY	REPRESENTATIVE PAIRS & COSINE SIMILARITY	SAFETY UI & EVALUATION RULE
Dangerous vs. Harmless (9 Pairs)	*B. caeruleus* vs *L. aulicus* (0.842, rank 4) · *E. carinatus* vs *B. trigonata* (0.799, rank 3) · *D. russelii* vs *P. molurus* (0.782, rank 3)	Track pair-wise false-negative rate (calling WHO Cat 1/2 harmless) with 10× asymmetric loss penalty
Venomous vs. Unlisted Congener	*B. niger* → *B. lividus* (0.908) · *C. malabaricus* → *C. anamallensis* (0.902) · *T. erythrurus* → *T. salazar* (0.897)	Never label "WHO Not Listed" as "Non-Venomous"; use family/genus heads to flag all Elapids & Vipers as venomous

100% Train Mask Coverage (129,241 Photos): 75.0% Clean Unflagged & 5,843 Hand Cutouts



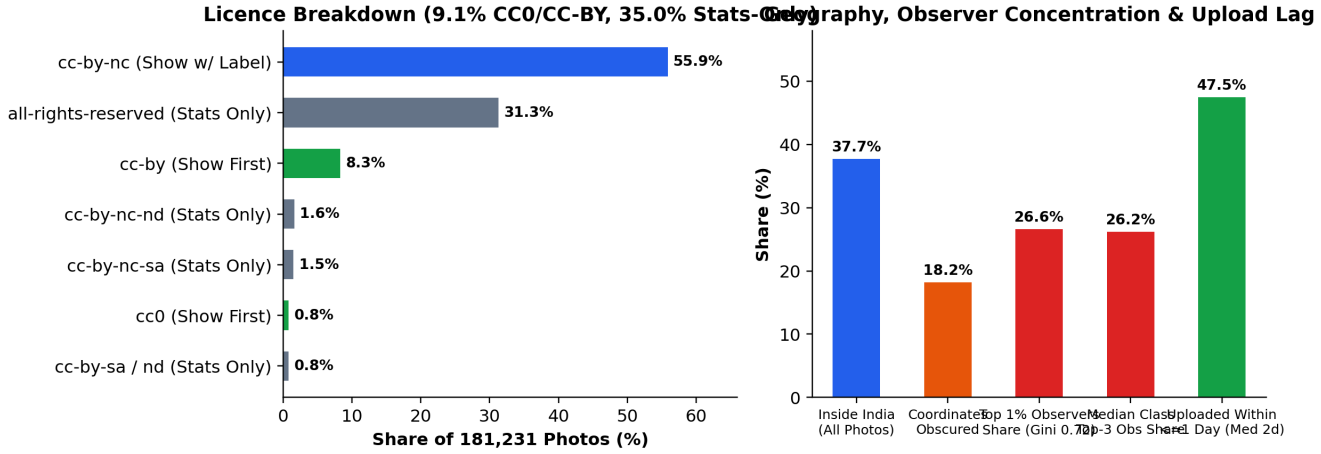


SAM Mask Quality & Thresholds

- **Complete Training Coverage**: `mask_stats.csv` covers **100.0% of `train`** (129,241 photos across all 208 classes: 9,742 WHO Cat 1; 6,579 WHO Cat 2; 112,920 Not Listed).
- **Exact Flagging Rule (100% Verified)**: **96,929 masks (75.0%)** pass all quality gates (`det_score`: 0.40–0.971, `mask_area_fraction`: 0.01–0.60, `fill_ratio`: 0.10–0.95).
- **Hand Removal & Multi-Snake Handling**: Detector automatically cut out human hands/gloves in **5,843 photos (4.5%)**; **3,932 photos (3.0%)** had no snake box (mostly thin *Ahaetulla* vine snakes).

SAM MASK SUBSET	PHOTO COUNT (SHARE)	HOW WE USE IT IN MILESTONE 3 TRAINING
Unflagged Clean Masks	**96,929 (75.0%)** · WHO Cat 1: 78.8% clean, Cat 2: 81.9% clean	Train segmentation head & mask-guided foreground cropping strictly on unflagged masks
Flagged Masks (32,312 = 25.0%)	Low score (13,121) · Low fill (10,759) · No snake (3,932) · Tiny area (3,368) · Huge/High fill (1,132)	Fallback to full-frame classification without mask suppression so thin arboreal snakes aren't dropped

37.7% India Photos, Monsoon Peak, Observer Gini = 0.72 & Strict Licence Hygiene



Real-World Context & Governance

- **Geography & Monsoon Seasonality**: **37.7%** of all photos are inside India (Western Ghats, NE & urban hubs) and **18.2%** have obscured coordinates. Sightings peak sharply during the **mid-year monsoon** across all WHO categories.
- **Observer Concentration (25,041 Photographers)**: Gini = **0.72**; the top 1% of observers took **26.6%** of all photos (max 2,189 by one user). In small classes, top-3 observers supply up to **80%–100%** of training photos.
- **Licence Compliance**: Only **9.1%** are `CC0/CC-BY` (24 classes have 0), **55.9%** are `CC-BY-NC`, and **35.0%** are statistics-only (`ARR / ND / SA`).

GOVERNANCE / CONTEXT DIMENSION	KEY NUMBER FROM NOTEBOOK	OPERATIONAL RULE FOR SERPENTGUARD AI
Observer Concentration per Class	Median 26.2% of a class's train photos come from its top-3 photographers	Cap maximum training photos per observer per class during epoch sampling to avoid learning camera style
Licence & Redistribution Policy	9.1% Show-First (CC0/CC-BY) · 55.9% Non-Commercial · 35.0% Stats-Only	Never redistribute raw ARR/ND images; ship only trained model weights & credit CC photographers in reports

What Is Completed in Milestone 2 vs. Next in Milestone 3 & Group 5 Team

PIPELINE STAGE	STATUS	CONCRETE ENGINEERING ACTION DERIVED FROM EDA
1–4. Hygiene, Format & Exact Dedup	Done in M2	33-col metadata + licence tracking, 500 px JPEG standard, 994 excluded + 272 cross-split pHash removed (**86/86 checks PASS**)
5–8. Splits, Labels & Train Norms	Done in M2	Observer-grouped splits (0 leakage), Cleanlab ranking, custom train RGB norms `[0.456, 0.439, 0.375]`; review 19 ≤2-bit near-copies
9. SAM Mask Crop & 224 vs 384 Input	Next · M3	Use **96,929 unflagged SAM masks** for foreground cropping + aspect-preserving letterbox; benchmark 224 vs 384 px for small vipers
10. Tail & Style Augmentation	Next · M3	Apply 90°/portrait transforms (for 56% portrait test), brightness/blur jitter (for darker `test_extra` & krait/wolf-snake sharpness gap)
11. Multi-Head & Safety-Aware Loss	Next · M3	Joint **Species (208) + Genus (77) + Family (14) + Venom Risk** heads, $1/N_c^{0.5}$ sampler, asymmetric WHO recall & unknown_pool abstention

Group 5 Project Team — IIT Madras BS in Data Science & AI Lab (2026 T3)

JAYESH BANSAL · TEAM LEAD	VIKAS · TEAM MEMBER	PUNEET BAJAJ · TEAM MEMBER	MANAS RASTOGI · TEAM MEMBER
Project Architecture, Problem Formulation & Safety Evaluation	Taxonomic & WHO Clinical Risk Analysis, Lookalike & Literature Audit	iNaturalist Data Engineering, SAM Masks Pipeline & Kaggle Datasets	Exploratory Data Analysis, DINOv2 / Cleanlab Audit & Quality Checks